Native F4/F5, F6 and root solvers

Run 20261019-174306-boolean-final-native-c19491, source 71c4c99d85. Dedicated c7i.2xlarge, one worker and a reserved complete physical core; four-worker controls reserve two cores/SMT siblings. Exact source/compiler/CPU/ISA and binary manifests remain in the raw tree. Baseline: frozen a78 algorithm with all three previously qualified GF2 switches. Candidate: identity-bound prepared construction and bounded quadratic-tail policies.

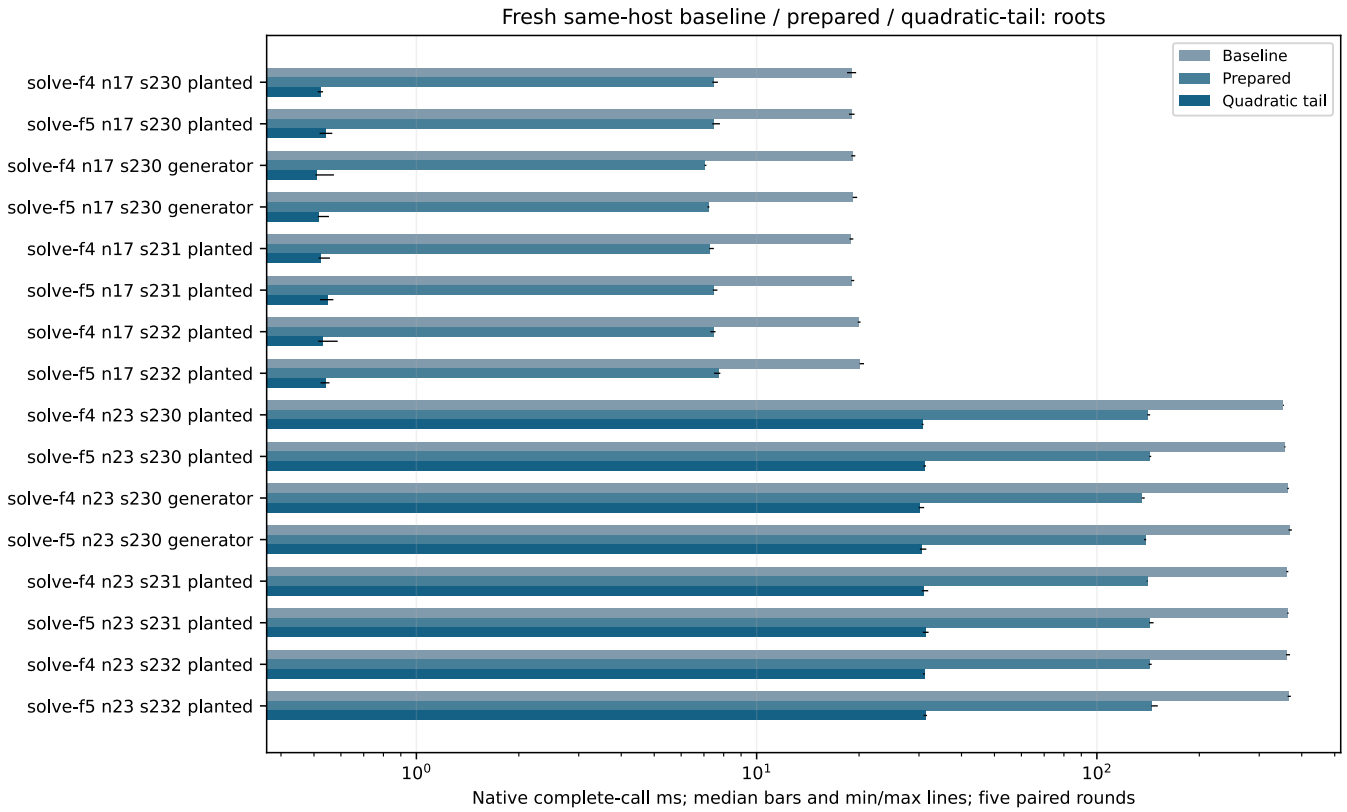
Original n17/n23 fixtures have16/22 Boolean variables on icv1-f2m17-tm101-00378d4e / icv1-f2m23-tm5197-1f85e9e1. Planted seed230 and fixed generator control plus planted holdouts231/232; random=true means generator here. n19's36-variable admission gap remains separate. No fixture or baseline is replaced by the full-DLP standard subspaces.

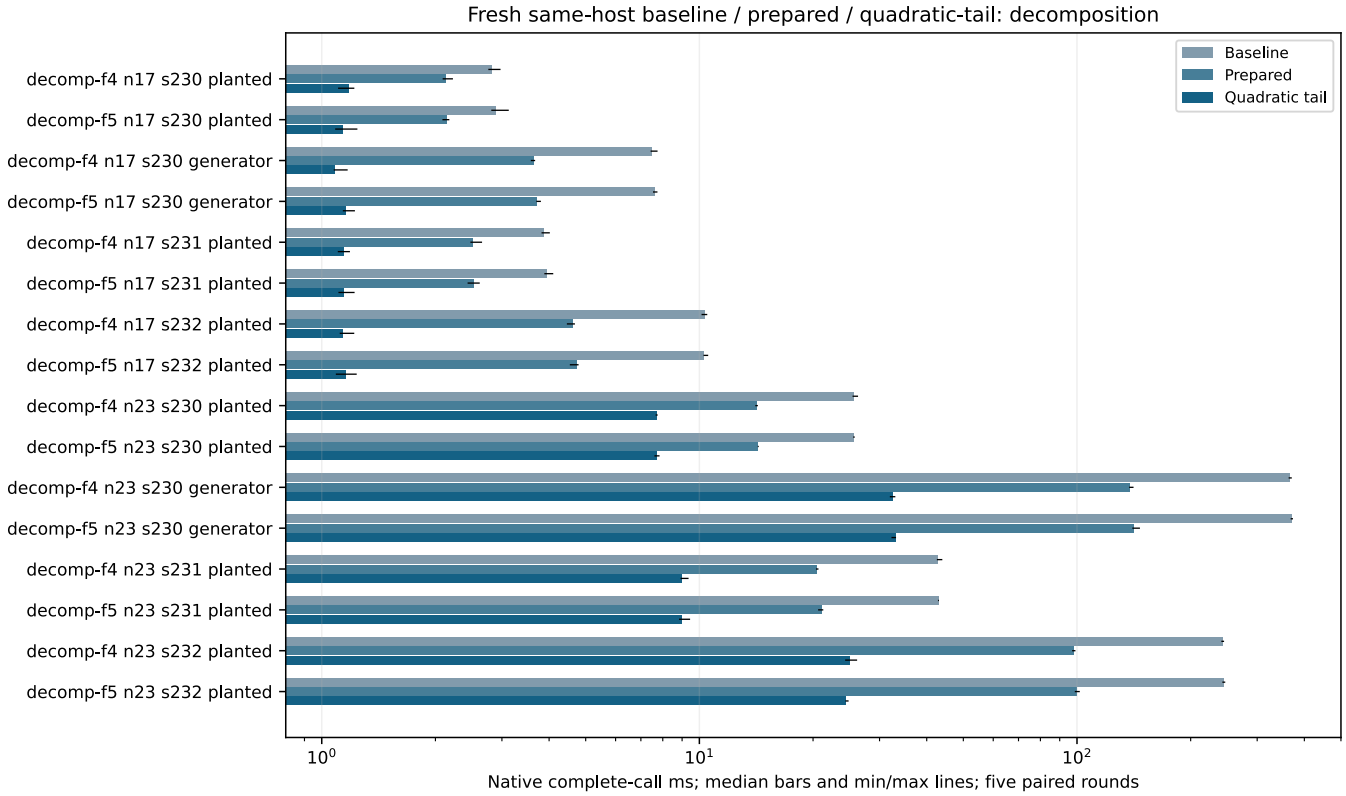
All eighteen arms are rerun with this source's declared policies. Complete original roots and first-witness decomposition remain separate; common curve/base construction is outside component timings. Fresh target ANF, solving, lifting and internal verification are charged to decomposition; Semaev fresh table setup is charged once. Independent original-root/group audits run after elapsed capture.

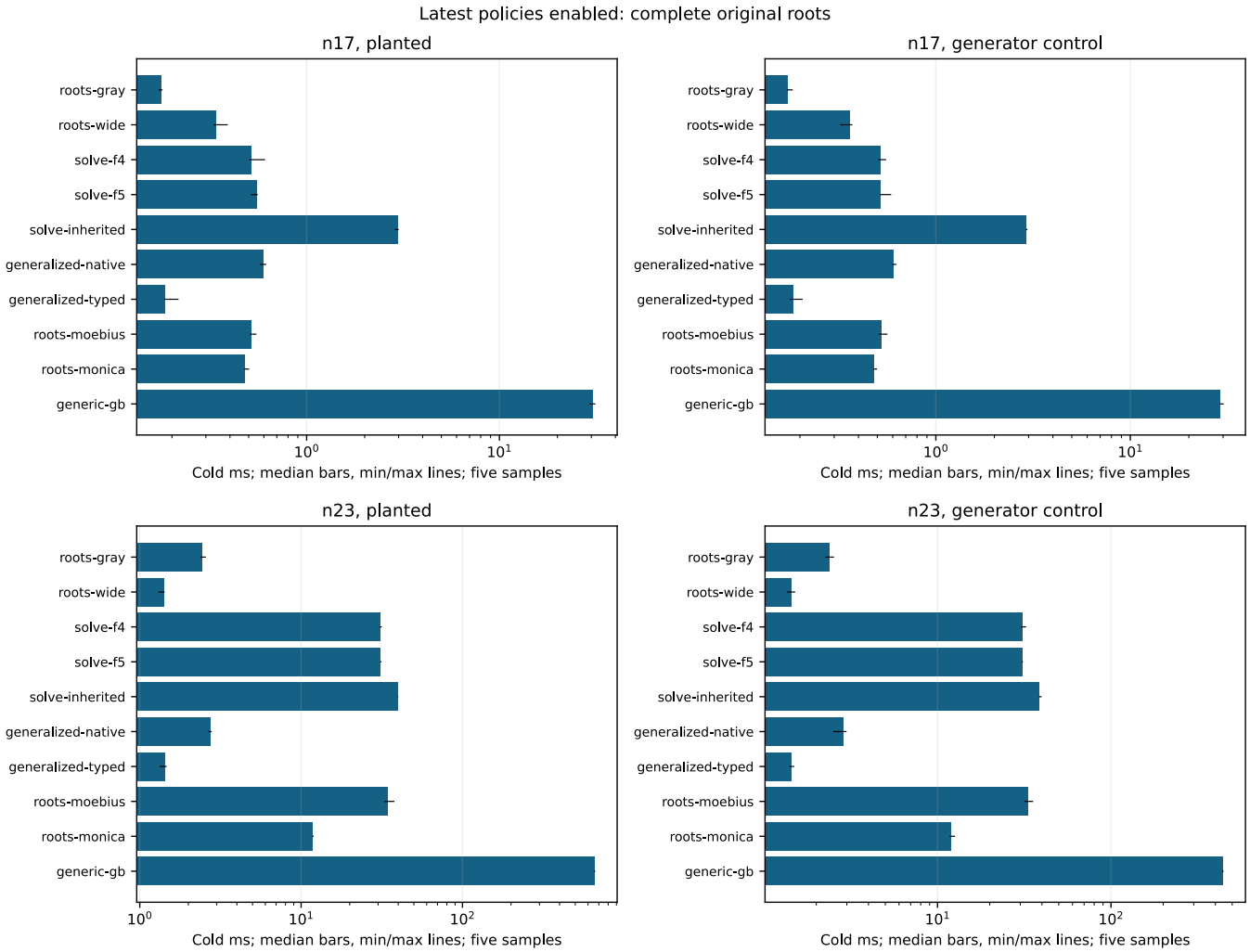
Five paired rounds, first eligible of at most three attempts, exact original outputs, exact3125-resample CI and max(2%, twice A/A noise) gate. Rejected attempts retained: 42. All four multiworker controls neutral: False. Every slower/neutral/missing cell remains in native-summary.json.

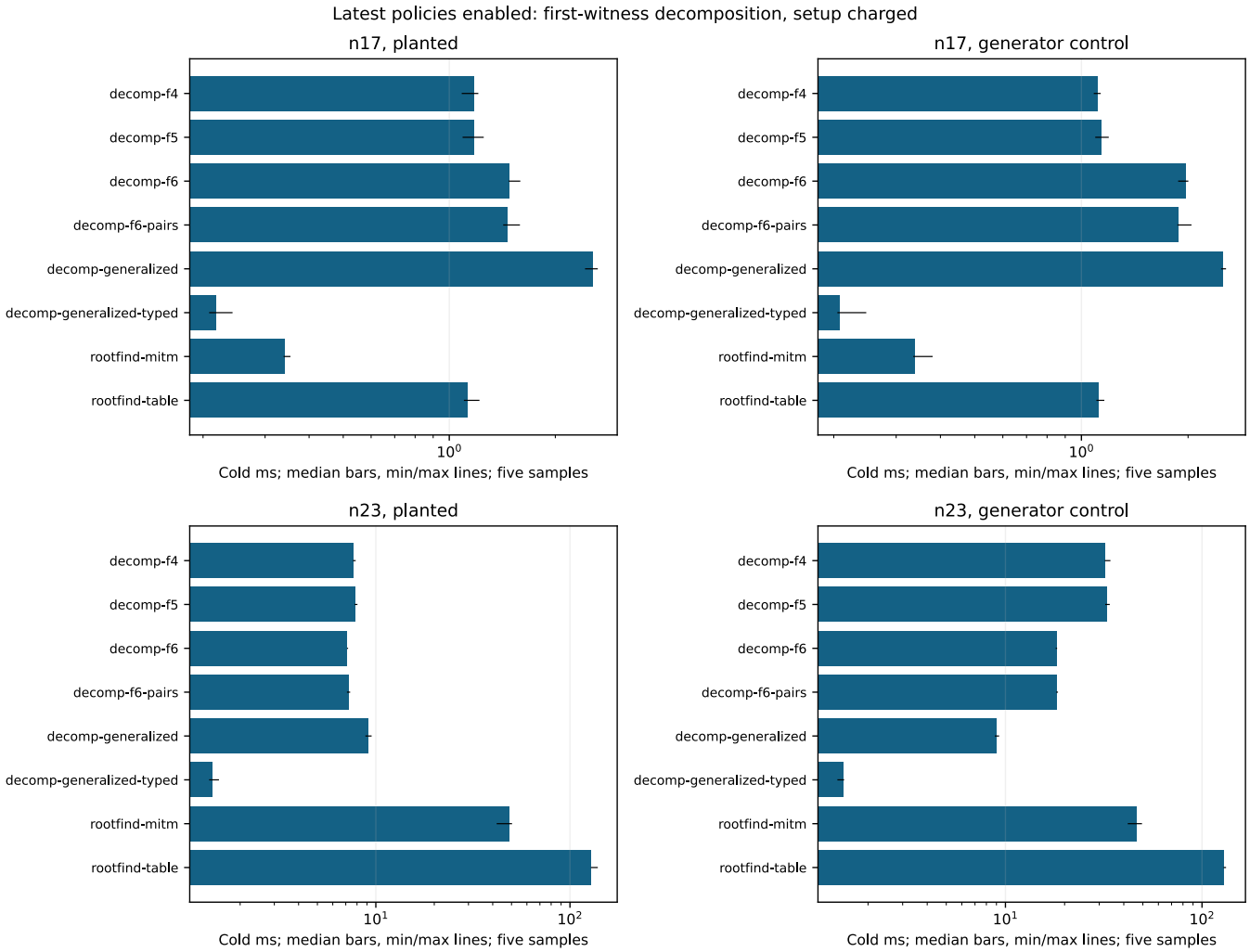
Prepared is a combined active-layout/top-degree/linear-tail/direct-packing ablation. Quadratic-tail retains an algebraic root reduction, then exhaustively completes at most16 free quadratic variables using the existing Gray kernel; original coordinates, root order, caps and verification remain enforced. This is a hybrid engineering change, not a novel enumeration algorithm. Other split rules/custom node oracles keep recursion.

The matrix switches stay opt-in. Source/protocols: 71c4c99d85, PROTOCOL.md and the frozen runner/protocols in the raw tree. Sources0acfc55b7f,71c4c99d85 and9bed104b0a are separate experiments. The final adaptive variants preserve cheap degree-two refutations below16 original variables; legacy policy names remain for replay. Full-table setup uses the same exact sorted tuples serially under an explicit one-worker resource contract. No cross-host pooling, external-fastest, Arm64/GPU, whole-IC/rho or hard-family8x conclusion follows from component calls.




Fresh same-host baseline / prepared / quadratic-tail: decomposition
Baseline
Prepared
Quadratic tail
decomp-f4 n17 s230 planted
decomp-f5 n17 s230 planted
decomp-f4 n17 s230 generator
decomp-f5 n17 s230 generator
decomp-f4 n17 s231 planted
decomp-f5 n17 s231 planted
decomp-f4 n17 s232 planted
decomp-f5 n17 s232 planted
decomp-f4 n23 s230 planted
decomp-f5 n23 s230 planted
decomp-f4 n23 s230 generator
decomp-f5 n23 s230 generator
decomp-f4 n23 s231 planted
decomp-f5 n23 s231 planted
decomp-f4 n23 s232 planted
decomp-f5 n23 s232 planted
10^0
10^1
10^2
Native complete-call ms; median bars and min/max lines; five paired rounds


Latest policies enabled: complete original roots
n17, planted
n17, generator control
n23, planted
n23, generator control
roots-gray
roots-wide
solve-f4
solve-f5
solve-inherited
generalized-native
generalized-typed
roots-moebius
roots-monica
generic-gb
Cold ms; median bars, min/max lines; five samples


Latest policies enabled: first-witness decomposition, setup charged
n17, planted
n17, generator control
n23, planted
n23, generator control
decomp-f4
decomp-f5
decomp-f6
decomp-f6-pairs
decomp-generalized
decomp-generalized-typed
rootfind-mitm
rootfind-table
10^0
10^1
10^2
Cold ms; median bars, min/max lines; five samples

All paired recursive-solver outcomes

Arm	n	Seed	Target	Before ms	Candidate ms	Ratio	Decision
solve-f4	17	230	P	19.0433	0.5230	36.4262	faster
solve-f5	17	230	P	19.0645	0.5410	34.9293	faster
decomp-f4	17	230	P	2.8103	1.1800	2.4125	faster
decomp-f5	17	230	P	2.8951	1.1361	2.5483	faster
solve-f4	17	230	G	19.2129	0.5104	37.5709	faster
solve-f5	17	230	G	19.1774	0.5172	37.3063	faster
decomp-f4	17	230	G	7.4807	1.0807	6.9349	faster
decomp-f5	17	230	G	7.6357	1.1576	6.6283	faster
solve-f4	17	231	P	18.9787	0.5221	36.0089	faster
solve-f5	17	231	P	19.0205	0.5485	35.2860	faster
decomp-f4	17	231	P	3.8810	1.1438	3.4884	faster
decomp-f5	17	231	P	3.9436	1.1421	3.5078	faster
solve-f4	17	232	P	20.0080	0.5309	37.3349	faster
solve-f5	17	232	P	20.0931	0.5408	37.3194	faster
decomp-f4	17	232	P	10.3217	1.1396	8.8925	faster
decomp-f5	17	232	P	10.2864	1.1557	8.8741	faster
solve-f4	23	230	P	353.0631	30.7438	11.5141	faster
solve-f5	23	230	P	356.0642	31.2119	11.4176	faster
decomp-f4	23	230	P	25.6580	7.6948	3.3291	faster
decomp-f5	23	230	P	25.6280	7.7112	3.3250	faster
solve-f4	23	230	G	364.4468	30.2591	12.0442	faster
solve-f5	23	230	G	370.1105	30.5390	12.1193	faster
decomp-f4	23	230	G	365.5542	32.5337	11.2291	faster
decomp-f5	23	230	G	369.6305	33.0418	11.2503	faster
solve-f4	23	231	P	362.3945	30.9733	11.6411	faster
solve-f5	23	231	P	364.1260	31.3525	11.6387	faster
decomp-f4	23	231	P	42.8421	8.9794	4.7591	faster
decomp-f5	23	231	P	43.0150	8.9591	4.8093	faster
solve-f4	23	232	P	362.7818	31.1121	11.7160	faster
solve-f5	23	232	P	365.5863	31.3206	11.7610	faster
decomp-f4	23	232	P	242.9192	24.9913	9.6493	faster
decomp-f5	23	232	P	244.2206	24.4465	10.0022	faster

Latest native complete-call ms (sample count)

Boundary	Method	n17 planted	n17 generator	n23 planted	n23 generator
complete_roots	roots-gray	0.1758 (5)	0.1727 (5)	2.4166 (5)	2.3692 (5)
complete_roots	roots-wide	0.3383 (5)	0.3618 (5)	1.4033 (5)	1.4385 (5)
complete_roots	solve-f4	0.5168 (5)	0.5172 (5)	30.9280 (5)	30.6685 (5)
complete_roots	solve-f5	0.5513 (5)	0.5200 (5)	31.0546 (5)	30.8427 (5)
complete_roots	solve-inherited	2.9738 (5)	2.9246 (5)	39.8591 (5)	38.5066 (5)
complete_roots	generalized-native	0.5937 (5)	0.6032 (5)	2.7420 (5)	2.8431 (5)
complete_roots	generalized-typed	0.1845 (5)	0.1851 (5)	1.4317 (5)	1.4242 (5)
complete_roots	roots-moebius	0.5189 (5)	0.5265 (5)	34.3487 (5)	32.9940 (5)
complete_roots	roots-monica	0.4784 (5)	0.4815 (5)	11.7774 (5)	11.9060 (5)
complete_roots	generic-gb	30.4903 (5)	28.8389 (5)	655.0972 (5)	440.5415 (5)
decomposition	decomp-f4	1.1755 (5)	1.1130 (5)	7.6494 (5)	32.1721 (5)
decomposition	decomp-f5	1.1749 (5)	1.1368 (5)	7.8036 (5)	32.6622 (5)
decomposition	decomp-f6	1.4770 (5)	1.9750 (5)	7.1248 (5)	18.1051 (5)
decomposition	decomp-f6-pairs	1.4596 (5)	1.8809 (5)	7.2238 (5)	18.1888 (5)
decomposition	decomp-generalized	2.5516 (5)	2.5056 (5)	9.1352 (5)	8.9492 (5)
decomposition	decomp-generalized-typed	0.2171 (5)	0.2081 (5)	1.4367 (5)	1.4867 (5)
decomposition	rootfind-mitm	0.3412 (5)	0.3395 (5)	48.5848 (5)	46.5130 (5)
decomposition	rootfind-table	1.1235 (5)	1.1163 (5)	128.4924 (5)	129.2967 (5)

Four-worker matrix controls

Arm	n	Before ms	After ms	Ratio	Decision
matrix-f4	17	7.7348	7.5342	1.0266	neutral
matrix-f5	17	8.0655	8.0306	1.0335	neutral
matrix-f4	23	59.4430	61.9066	0.9633	slower
matrix-f5	23	59.4071	57.8192	1.0010	neutral

Instrumented matrix phase medians, ms; separate nonadditive observations

Arm	n	Target	Policy	Build ms	Reduce ms	Readback ms	Calls
solve-f4	17	P	baseline	8.8756	9.1566	0.3874	126.0
solve-f4	17	P	prepared	2.5933	4.5430	0.0327	63.0
solve-f4	17	P	quadratic-tail	0.2818	0.1263	0.0003	1.0
solve-f5	17	P	baseline	9.0522	9.0178	0.3794	126.0
solve-f5	17	P	prepared	2.7632	4.4632	0.0284	63.0
solve-f5	17	P	quadratic-tail	0.2983	0.1259	0.0002	1.0
decomp-f4	17	P	baseline	1.0943	0.9383	0.0395	12.0
decomp-f4	17	P	prepared	0.8315	0.5480	0.0020	6.0
decomp-f4	17	P	quadratic-tail	0.2281	0.1265	0.0003	1.0
decomp-f5	17	P	baseline	1.1079	0.9476	0.0393	12.0
decomp-f5	17	P	prepared	0.8489	0.5313	0.0017	6.0
decomp-f5	17	P	quadratic-tail	0.2289	0.1237	0.0002	1.0
solve-f4	17	G	baseline	9.1485	9.1817	0.3733	126.0
solve-f4	17	G	prepared	2.6173	4.1115	0.0272	63.0
solve-f4	17	G	quadratic-tail	0.2807	0.1141	0.0003	1.0
solve-f5	17	G	baseline	9.2479	9.0682	0.3728	126.0
solve-f5	17	G	prepared	2.7807	4.1329	0.0240	63.0
solve-f5	17	G	quadratic-tail	0.2867	0.1129	0.0003	1.0
decomp-f4	17	G	baseline	3.2190	3.2356	0.1311	44.0
decomp-f4	17	G	prepared	1.3047	1.5018	0.0085	22.0
decomp-f4	17	G	quadratic-tail	0.2178	0.1108	0.0003	1.0
decomp-f5	17	G	baseline	3.3063	3.2220	0.1311	44.0
decomp-f5	17	G	prepared	1.3687	1.5260	0.0080	22.0
decomp-f5	17	G	quadratic-tail	0.2270	0.1124	0.0002	1.0
solve-f4	23	P	baseline	154.2482	185.5118	5.8229	1022.0
solve-f4	23	P	prepared	37.9914	100.1398	0.2870	511.0
solve-f4	23	P	quadratic-tail	10.4035	15.9620	0.0060	63.0
solve-f5	23	P	baseline	157.6023	185.3282	5.8600	1022.0
solve-f5	23	P	prepared	39.3776	100.4474	0.2741	511.0
solve-f5	23	P	quadratic-tail	10.7040	16.1410	0.0041	63.0
decomp-f4	23	P	baseline	10.8990	11.8709	0.3695	64.0
decomp-f4	23	P	prepared	4.8868	6.9019	0.0191	32.0
decomp-f4	23	P	quadratic-tail	2.9993	2.2132	0.0014	7.0
decomp-f5	23	P	baseline	10.9725	11.7113	0.3691	64.0
decomp-f5	23	P	prepared	4.9918	6.9335	0.0173	32.0
decomp-f5	23	P	quadratic-tail	3.0309	2.2381	0.0007	7.0
solve-f4	23	G	baseline	161.5615	188.6295	5.9307	1022.0
solve-f4	23	G	prepared	38.9588	93.4667	0.2582	511.0
solve-f4	23	G	quadratic-tail	10.6806	15.1731	0.0055	63.0
solve-f5	23	G	baseline	166.4366	190.6396	5.9664	1022.0
solve-f5	23	G	prepared	41.3334	94.3479	0.2541	511.0
solve-f5	23	G	quadratic-tail	11.0164	15.1026	0.0033	63.0
decomp-f4	23	G	baseline	161.6477	188.7095	5.9108	1022.0
decomp-f4	23	G	prepared	39.0343	93.8238	0.2510	511.0
decomp-f4	23	G	quadratic-tail	10.4695	15.3318	0.0034	63.0
decomp-f5	23	G	baseline	165.9340	188.4201	5.8962	1022.0
decomp-f5	23	G	prepared	40.9341	95.0892	0.2610	511.0
decomp-f5	23	G	quadratic-tail	10.8517	15.3872	0.0034	63.0